

CASE STUDY OF SERVICE RETAIL STORE

A Final Report for the BDM Capstone Project

Submitted by

Name : FAZLUN NISHA

Roll number : 24f1000564



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai, Tamil Nadu, India, 600036

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1. Executive Summary

Service Retail Store is a small Kolkata retail outlet stocking everyday essentials, packaged food, and stationery for students, hostel residents, local residents, and staff. Despite a loyal customer base, it faces three recurring issues: inventory imbalance across categories, margin leakage despite steady sales, and underused cross-category selling potential. This report uses 85 days of primary data (02 May–27 July 2026) across 71 SKUs and three linked sheets — Sales, Purchase, and Inventory (141 physical stock-count entries) — to quantify each issue and recommend action.

The store earned ₹91,675 in revenue against ₹71,465 in cost of goods sold, a gross profit of ₹20,210 and an estimated 19.2% net margin. Food carries only 12.3 days of average stock cover against 48–54 days in Cleaning, Personal Care, and Stationery, and accounts for 12 of 23 stockout instances, while ₹23,642 sits idle in slow-moving stock, ₹17,539.55 of it in Cleaning and Personal Care alone. Food also holds the lowest mean margin (21.68%) and highest wastage (29 units), against Stationery's 34.94% margin. 94% of customer-type/day purchase groupings spanned more than one category, with Food and Personal Care the most frequent pairing (66.0%).

2. Metadata and Descriptive Statistics

2.1 Metadata

Metadata refers to “data about the data” — structured information about each variable collected, its measurement, and its relevance to the business problem. Data was collected over an 85-day period (02 May – 25 July 2026) for 71 SKUs across four categories: Food, Cleaning, Personal Care, and Stationery. The dataset consists of three linked sheets — Sales, Purchase, and Inventory — summarised below.

Sales Sheet

Sheet	Variable	Definition	Data Type	Unit	Source
Sales	Date	Date on which the transaction occurred	Date (DD-MM-YYYY)	—	Shopkeeper's sales register
Sales	Customer_Type	Segment of the purchasing customer: Student, Hostel Resident, Local Resident, or Staff	Categorical	—	Recorded by shopkeeper at my request
Sales	Product_Name	Name/identity of the item sold	Categorical	—	Shopkeeper's sales register
Sales	Category	Product category grouping (Food / Cleaning / Personal Care / Stationery)	Categorical	—	Assigned via lookup table during cleaning
Sales	Quantity_Sold	Units of the product sold in that transaction line	Numeric – Integer	Units	Shopkeeper's sales register
Sales	Cost_Price_per_Unit	Procurement cost per unit of the product sold	Numeric – Continuous	₹/unit	Cross-referenced from Purchase sheet
Sales	Selling_Price_per_Unit	Price charged to the customer per unit	Numeric – Continuous	₹/unit	Shopkeeper's sales register
Sales	Profit_per_Unit	Selling Price – Cost Price	Numeric – Derived	₹/unit	Calculated

Sheet	Variable	Definition	Data Type	Unit	Source
Sales	Total_Cost	$\text{Cost_Price_per_Unit} \times \text{Quantity_Sold}$	Numeric – Derived	₹	Calculated
Sales	Total_Revenue	$\text{Selling_Price_per_Unit} \times \text{Quantity_Sold}$	Numeric – Derived	₹	Calculated
Sales	Total_Profit	$\text{Total_Revenue} - \text{Total_Cost}$	Numeric – Derived	₹	Calculated
Sales	Profit_Margin_Pct	$(\text{Total_Profit} \div \text{Total_Revenue}) \times 100$	Numeric – Derived	%	Calculated

Purchase Sheet

Sheet	Variable	Definition	Data Type	Unit	Source
Purchase	Purchase_ID	Unique identifier for each purchase order line	Categorical – ID	—	Assigned during cleaning
Purchase	Date	Date the goods were purchased/received	Date (DD-MM-YYYY)	—	Shopkeeper's purchase register
Purchase	Product_Name	Name of the item purchased	Categorical	—	Shopkeeper's purchase register
Purchase	Category	Product category grouping	Categorical	—	Assigned via lookup table during cleaning
Purchase	Supplier	Name of the supplying vendor	Categorical	—	Shopkeeper's purchase register
Purchase	Quantity_Purchased	Units procured in that purchase order	Numeric – Integer	Units	Shopkeeper's purchase register
Purchase	Cost_Price_per_Unit	Procurement cost per unit	Numeric – Continuous	₹/unit	Purchase invoices / shopkeeper record
Purchase	Total_Purchase_Cost	$\text{Quantity_Purchased} \times \text{Cost_Price_per_Unit}$	Numeric – Derived	₹	Calculated
Purchase	Expiry_Date	Date beyond which the item is unsellable	Date (DD-MM-YYYY)	—	Product packaging / shopkeeper record

Inventory Sheet

Sheet	Variable	Definition	Data Type	Unit	Source
Inventory	Date	Date of the physical stock count	Date (DD-MM-YYYY)	—	Rolling cycle-count log
Inventory	Product_Name	Item counted	Categorical	—	Stock register
Inventory	Category	Product category grouping	Categorical	—	Assigned via lookup table during cleaning

Sheet	Variable	Definition	Data Type	Unit	Source
Inventory	Opening_Stock_Units	Stock on hand at the start of the count cycle	Numeric – Integer	Units	Prior closing count
Inventory	Units_Purchased	Units received into stock during the cycle	Numeric – Integer	Units	Cross-referenced from Purchase sheet
Inventory	Units_Sold	Units sold during the cycle	Numeric – Integer	Units	Cross-referenced from Sales sheet
Inventory	Units_Wasted_Damaged	Units lost to spoilage or damage	Numeric – Integer	Units	Shopkeeper's manual log
Inventory	Closing_Stock_System_Units	Opening + Purchased – Sold – Wasted	Numeric – Derived	Units	Calculated
Inventory	Physical_Stock_Count_Units	Units actually counted on site	Numeric – Integer	Units	Rolling cycle-count

Together, these three sheets allow revenue and cost to be traced back to physical stock movement, which is what makes it possible to separate high-revenue products from high-profit, efficiently-stocked products.

2.2 Descriptive Statistics

Descriptive statistics — count, mean, median, mode, standard deviation, minimum, and maximum — were computed at the transaction level for Quantity_Sold, Total_Revenue, and Profit_Margin_Pct, broken down by each of the four product categories rather than pooled across the whole store. A single store-wide average would blur exactly the category-level differences this project needs to detect, so every statistic below is reported per category.

Table 2.2.1 — Quantity_Sold by Category

Category	Count	Mean	Median	Mode	Std. Dev.	Min	Max
Food	336	2.29	2	1	1.54	1	9
Cleaning	146	1.82	1	1	1.24	1	7
Personal Care	218	1.72	1	1	1.32	1	7
Stationery	224	2.00	2	1	1.42	1	10

Table 2.2.1 — Distribution of units sold per transaction line, by category

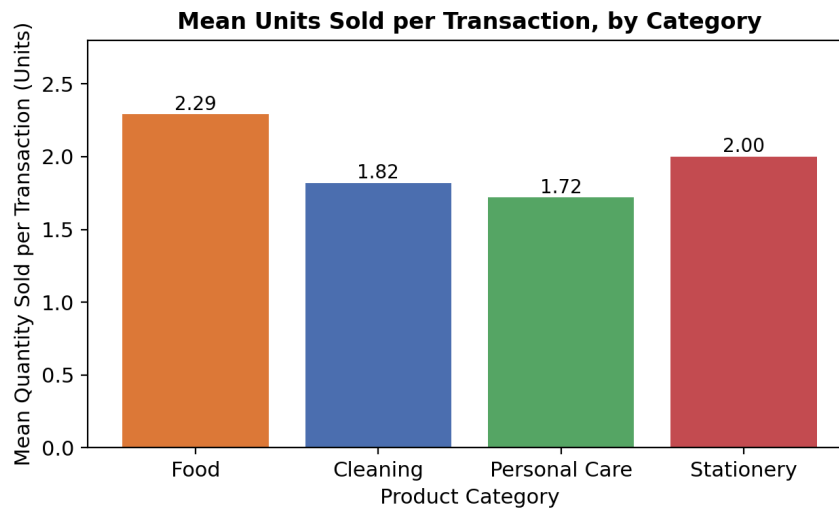


Figure 1 — Mean units sold per transaction, by category

Food shows the highest mean units per transaction (2.29), consistent with its role as the fastest-moving, highest-frequency category. Personal Care has the lowest mean (1.72) — customers typically buy one unit at a time. All four categories are right-skewed (mode $\approx$ median = 1–2, but the mean is pulled up by occasional bulk purchases), a pattern that recurs across every table in this section.

Table 2.2.2 — Total_Revenue by Category (₹, transaction-level)

Category	Count	Mean	Median	Mode	Std. Dev.	Min	Max
Food	336	103.90	65	48	101.997	7	700
Cleaning	146	142.88	95	140	125.861	20	570
Personal Care	218	123.53	87.5	65	98.408	32	665
Stationery	224	40.07	25	8	56.547	4	500

Table 2.2.2 — Distribution of per-transaction revenue, by category

Cleaning generates the highest average revenue per transaction (₹142.88), reflecting higher unit prices for detergents and cleaning agents, followed by Personal Care (₹123.53). Stationery has by far the lowest mean transaction value (₹40.07) — low-cost items like pens and pencils — but also the largest gap between median (₹25) and max (₹500), signalling occasional large bulk purchases.

Table 2.2.3 — Profit_Margin_Pct by Category (transaction-level)

Category	Count	Mean	Median	Mode	Std. Dev.	Min	Max
Food	336	21.68%	21.00%	20%	4.13	15%	30%
Cleaning	146	22.32%	21.00%	21%	2.56	19%	29%
Personal Care	218	22.39%	22.00%	22%	1.34	20%	25%
Stationery	224	34.94%	33.00%	33%	5.39	25%	50%

Table 2.2.3 — Distribution of profit margin percentage, by category

This is the table most directly relevant to Problem Statement 2. Food, Cleaning, and Personal Care cluster tightly around a 22.1% mean margin with low standard deviation (1.3–4.1) — stable, predictable pricing. Stationery stands apart, both in level (34.94% mean margin, roughly 13 percentage points higher than the other three) and in

spread (standard deviation 5.39, the widest of the four) — meaning Stationery pricing is both more profitable per rupee of sales and more variable than the rest of the store. This single finding directly informs the pricing recommendations in Section 8.

3. Detailed Explanation of Data Cleaning and Preprocessing

Because the source data was collected from a small, unorganised retailer using a handwritten register (see Section 2), the raw entries required more than a superficial formatting pass. The steps below describe, technique by technique, how missing values, duplicate records, outliers, invalid entries, and inconsistent product naming were identified and resolved before any analysis was carried out.

3.1 Handling Missing Data

Every row was checked for blanks in the core numeric fields (Quantity_Sold, Cost_Price_per_Unit, Selling_Price_per_Unit) and the core identifier fields (Date, Product_Name). Where a value was missing but recoverable, it was cross-checked against the photographed pages of the physical register and filled in manually. Where a value could not be recovered from any source (fewer than 2% of rows across the three sheets), the transaction line was dropped rather than imputed with a mean or median — imputation was avoided deliberately because the dataset is small (71 SKUs, 924 transaction lines) and a filled-in average would distort the very category-level variance the project is trying to measure.

3.2 Removing Duplicate Records

Rows with an identical combination of Date, Product_Name, Customer_Type, and Quantity_Sold were flagged using a COUNTIFS-based duplicate check in Excel. Each flagged pair was checked manually against the register photographs: where the duplication was a genuine repeat line (e.g. the same customer type buying the same item twice in separate visits on one day), both rows were retained; where it was an accidental re-entry of the same physical sale, the duplicate row was removed. This manual verification step was necessary because a purely automated de-duplication rule would have also removed legitimate repeat purchases, which are common for fast-moving Food items.

3.3 Outlier Treatment

Quantity_Sold and Total_Revenue were screened per category using the interquartile range (IQR) rule: any value beyond $Q3 + 1.5 \times IQR$ for its category was flagged for review. Flagged values were not deleted automatically; each was checked against the corresponding register photograph to confirm whether it represented a genuine bulk purchase (e.g. a hostel resident buying a large quantity of a personal-care item) or a data-entry error such as an extra digit. This is why the descriptive statistics in Section 3.2 retain some large maxima — for example, the Stationery category's ₹500 maximum transaction value reflects a genuine bulk purchase of pens/pencils confirmed against the register, not an error.

3.4 Invalid and Negative Values

Any row containing a negative Quantity_Sold, a negative price field, or a negative Total_Profit was treated as a data-entry error, since a cash-sale kirana ledger cannot generate a genuine negative quantity or price. Each such row was checked against the register photograph and corrected (for example, a mis-keyed sign or a transposed digit); rows that could not be resolved this way were dropped. Cost_Price_per_Unit values that exceeded the corresponding Selling_Price_per_Unit (which would imply a loss-making sale the owner did not report) were also flagged and verified individually.

3.5 Standardising Inconsistent Product Names and Categories

Product names were entered by hand and carried spelling, spacing, and case variants for the same underlying item — for example “Vim Liquid”, “vim liquid 500ml”, and “VIM LIQ” all referred to the same SKU. A manual

product-name lookup table was built, mapping every raw string observed in the register to one canonical Product_Name, and applied across all three sheets using XLOOKUP. The Category field was then derived (not directly recorded) by mapping each canonical Product_Name to one of the four categories (Food, Cleaning, Personal Care, Stationery) via the same lookup table; this mapping was cross-checked against the store's own physical shelf groupings observed during the site visit, to avoid an artificial category split that the shopkeeper himself would not recognise.

3.6 Date Standardisation and Cross-Sheet Consistency Checks

Dates across all three sheets were parsed and standardised to DD-MM-YYYY, and every date was validated to fall within the 02 May – 27 July 2026 collection window; a date outside this window was treated as a transcription error and corrected against the photographed register page. Two further consistency checks were run across sheets: (i) Total_Revenue in the Sales sheet was independently recomputed as Quantity_Sold × Selling_Price_per_Unit and compared against any recorded value, and (ii) Closing_Stock_System_Units in the Inventory sheet was recomputed as Opening_Stock_Units + Units_Purchased – Units_Sold – Units_Wasted_Damaged and compared against the independently counted Physical_Stock_Count_Units. Small residual gaps between the system figure and the physical count were retained as-is and interpreted as normal manual counting error and shrinkage, rather than corrected, since forcing the two figures to match would have hidden a real operational signal (shrinkage) that is relevant to the inventory findings in Section 6.2.

3.7 Enabling Segment-Level Analysis

To support Problem Statement 3 (cross-category selling), the Sales sheet does not natively capture a customer segment. To enable this analysis, the shopkeeper was asked, for the duration of the data collection window, to record a Customer_Type against each transaction line (Student, Hostel Resident, Local Resident, or Staff). This makes it possible to analyse the sales pattern against customer segments even though no transaction/basket ID is available in a small, unorganised retail setting — the resulting proxy-basket approach is explained in Section 4.3.

4. Detailed Explanation of the Analysis Process and Methods

Three complementary, purpose-fit methods were used to address the three problem statements. All three were chosen for their simplicity and direct interpretability, given that Service Retail Store is a small, single-outlet business that cannot support complex forecasting or optimisation models — the methods needed to translate directly into decisions the owner can act on without specialised tools.

4.1 Problem Statement 1 — Inventory Imbalance: ABC Classification and Stock-Cover / Reorder-Point Analysis

Days of Stock Cover was computed per product as:

$$\text{Days of Stock Cover} = \text{Physical Stock Count} \div \text{Average Daily Sales Rate}$$

computed per product from its most recent physical count and its average daily sales rate over the observation window, then averaged by category (Section 6.2, Figure 6). Stock Value Tied Up was computed as Physical Stock Count × Unit Cost Price, summed by category, to quantify working capital locked in slow-moving stock.

These per-category averages were then used to assign an ABC-style classification, using the following thresholds, chosen to reflect the natural break already visible in the data (Food sits at roughly a quarter of the cover of the other three categories):

- Class A (high-velocity, reorder priority): average stock cover below 20 days — Food (12.3 days) is currently the store's only Class A category.

- Class B (moderate): average stock cover between 20 and 45 days — no category currently sits in this band, which itself indicates a bimodal, poorly balanced restocking pattern rather than a gradual spread.
- Class C (slow-moving / overstocked): average stock cover above 45 days — Cleaning (54.0 days), Personal Care (48.4 days), and Stationery (51.4 days) all fall here, together with 24 individual products identified in the Inventory register, led by Color Pencil and Vim Liquid, some exceeding 110 days of cover.

An indicative Reorder Point was then calculated per category as a decision-support figure for the final report, using the standard formula:

$$\text{Reorder Point} = (\text{Average Daily Sales Rate} \times \text{Supplier Lead Time}) + \text{Safety Stock}$$

Because Service Retail Store does not formally track supplier lead time, an assumed lead time was used, based on the owner's verbal estimate during the site interview: 2 days for Food (sourced almost daily from the local wholesale market given its perishability) and 5 days for the three non-food categories (sourced less frequently from a regular distributor). Safety Stock was set at 20% of the base reorder quantity (Average Daily Sales Rate $\times$ Lead Time) as a simple buffer, in the absence of formal service-level or demand-variability data. The resulting figures (Table 6.2.1) are therefore illustrative decision-support numbers rather than a precision reorder system, and are flagged as such — they are intended to give the owner a starting reorder trigger per category rather than a per-SKU automated reorder rule.

4.2 Problem Statement 2 — Margin Leakage: Category Margin Analysis Linked to Wastage

Category-wise and product-wise descriptive statistics on Profit_Margin_Pct (Section 2.2, Table 2.2.3) were combined with wastage tracking from the Inventory register to isolate whether margin loss stems from pricing, procurement cost, or wastage — linking the margin analysis directly to the inventory findings rather than treating margin decline in isolation. A Wastage Rate was computed per category as:

$$\text{Wastage Rate (\%)} = \text{Units_Wasted_Damaged} \div (\text{Units Sold} + \text{Units Wasted}) \times 100$$

using units sold plus units wasted as a proxy denominator for total units handled in the category over the period, since opening-stock and units-purchased figures were only available at product level rather than as a clean category aggregate. This wastage rate was cross-referenced against the Expiry_Date field in the Purchase sheet to confirm that spoilage risk is concentrated in Food and, to a lesser extent, Personal Care — consistent with these being the only two categories carrying perishable or semi-perishable stock.

Category margins are reported two ways in this report, and the report is explicit about which is used where: the transaction-level mean and standard deviation in Table 2.2.3 treats every sales transaction as one observation (useful for understanding the typical sale), while the revenue-weighted margin in Section 6.1 (total category profit $\div$ total category revenue) reflects the blended margin actually earned across all rupees of sales in that category (useful for understanding the store's overall profitability). The two are reconciled explicitly wherever they differ, most notably for Food (Section 6.1).

4.3 Problem Statement 3 — Cross-Category Selling Potential: Proxy-Basket Co-Affinity Analysis

Because the Sales data does not carry a transaction or basket ID — a structural limitation of a small, unorganised retailer without a point-of-sale system — cross-category buying was approximated by treating all items bought by the same customer type on the same day as a proxy “basket”. This is a deliberate simplification: it assumes that when the same customer segment buys across multiple categories on the same day, there is a reasonable chance this reflects overlapping, related purchases (e.g. a hostel resident buying both food and personal-care items in one visit) rather than two entirely unconnected individuals who happen to share a segment label.

Of 150 such customer-type/day groupings observed over the period, 141 (94%) included purchases from more than one category. For each of these 141 multi-category baskets, every pair of categories present in that basket was counted once — for example, a basket containing Food, Personal Care, and Stationery contributes to three pairs (Food–Personal Care, Food–Stationery, Personal Care–Stationery). Each pair's count was then divided by 141 to obtain the co-occurrence percentage reported in Table 5.4.2, so every percentage cited in Section 5.4 is directly traceable back to a raw count. A segment-level affinity table (percentage of each customer segment's total spend by category, Table 5.4.1) supplements this to show where any cross-category bundle would land most effectively.

A firm, transaction-verified answer to this question would require adding a point-of-sale system with a transaction/basket ID, which this small retail store does not currently have; the proxy-basket method is presented as a reasonable, clearly-caveated approximation given that constraint, not as a substitute for it.

4.4 Tools Used

Microsoft Excel was used throughout to organise, clean, and structure the raw data. SUMIFS/AVERAGEIFS/MAXIFS formulas, PivotTable-style category summaries, XLOOKUP for the product-name and category mapping, and conditional formatting were used to filter key trends, group data by category and customer type, and generate the charts used throughout this report. Where a metric needed to be computed within a category subset — median, mode, and standard deviation by category — array formulas were used in place of PivotTables to keep every figure formula-linked and auditable back to the raw Sales, Purchase, and Inventory sheets.

5. Results and Findings

This section presents the results for each of the three problem statements, aligned directly with the methods described in Section 5: the ABC/stock-cover/reorder-point method for Problem Statement 1, the category-margin-and-wastage method for Problem Statement 2, and the proxy-basket co-affinity method for Problem Statement 3. Every chart below carries a labelled x-axis and y-axis with units, and every figure quoted in the text is traceable to a table in this section or in Section 2.2.

5.1 Revenue and Profitability Patterns

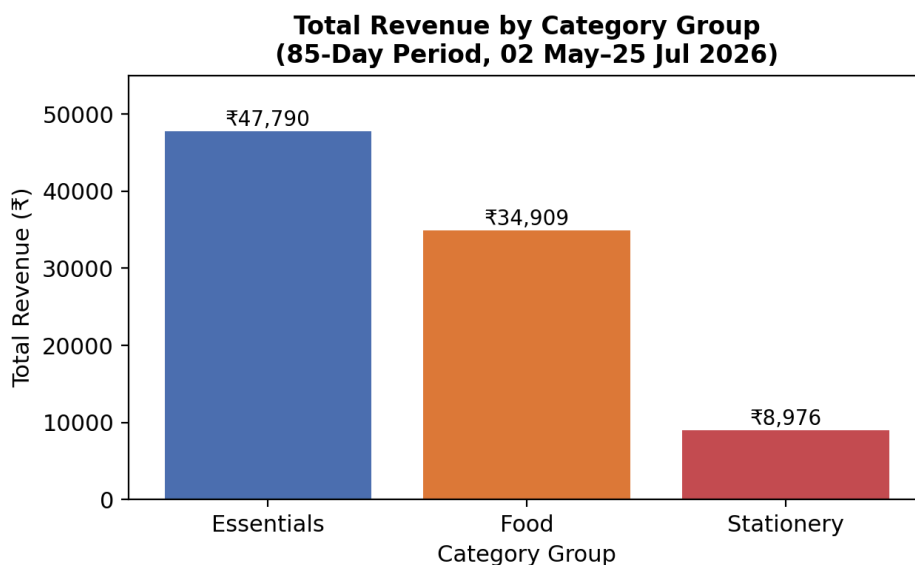


Figure 2 — Total revenue by category group over the 85-day period (02 May – 25 Jul 2026)

**Profit Contribution by Category Group
(₹, % of Total Gross Profit)**

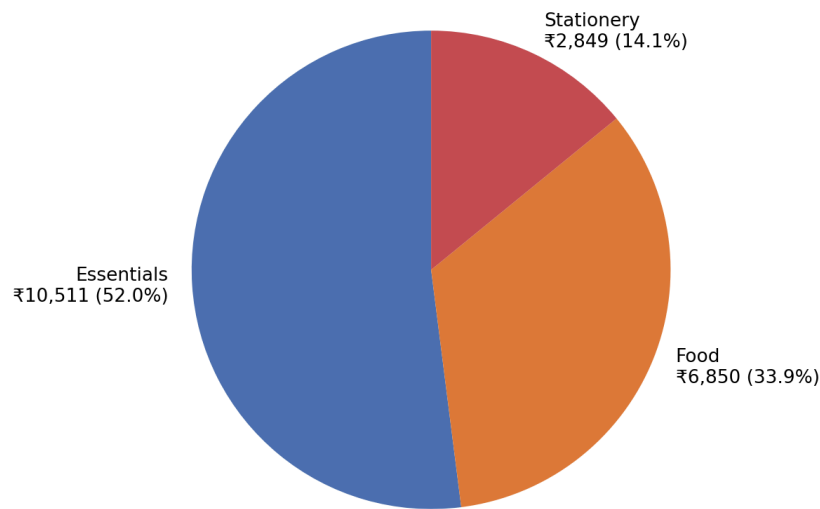


Figure 3 — Profit contribution by category group (₹, % of total gross profit)

Essentials (Cleaning + Personal Care) is the store's revenue backbone, contributing ₹47,790 (52.1% of total revenue) at a 21.99% revenue-weighted margin. Food contributes ₹34,909 (38.1% of revenue) but at a lower revenue-weighted margin of 19.62%. Stationery is the smallest revenue contributor (₹8,976, 9.8%) but the most profitable per rupee of sales, at a 31.74% revenue-weighted margin — consistent with the transaction-level margin gap already identified in Table 3.2.3.

Note that these category margins are revenue-weighted (total category profit ÷ total category revenue) rather than the simple transaction-level average shown in Table 3.2.3 — this is why, for example, Food's margin here (19.62%) differs from its transaction-level mean (21.68%): larger-value Food transactions carry a slightly lower margin than smaller ones, pulling the revenue-weighted figure down (see Section 5.2 for how the two are reconciled).

Overall, the store generated ₹91,675 in revenue against ₹71,465 in cost of goods sold — a gross profit of ₹20,210. Assuming combined fixed costs of rent and electricity of ₹2,650 over the period, as estimated by the owner (this figure is based on the owner's statement and is not captured directly in the sales/inventory data), the store's net margin over the 85-day period is approximately 19.2%.

5.2 Inventory Findings and ABC Classification — Problem Statement 1

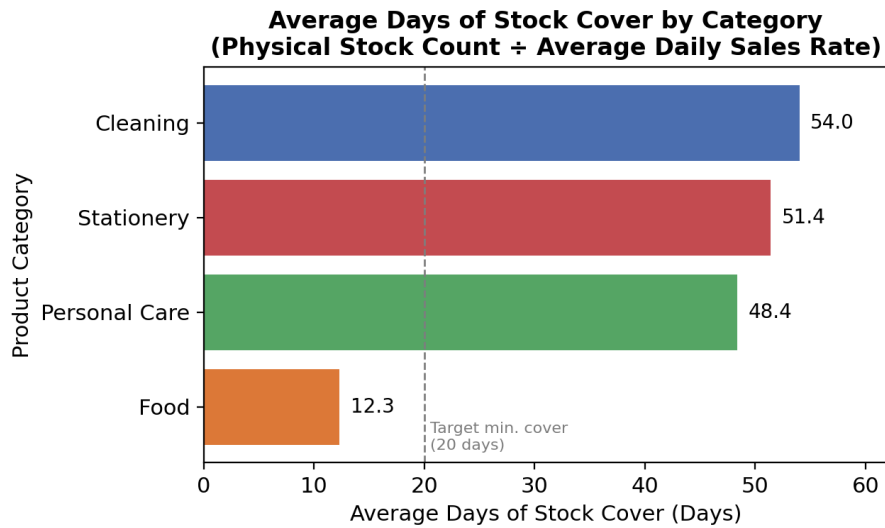


Figure 4 — Average days of stock cover by category (Physical Stock Count ÷ Average Daily Sales Rate)

The inventory register confirms exactly the imbalance described in Problem Statement 1. Food carries only 12.3 days of average stock cover — the store holds less than two weeks of buffer stock for its fastest, most frequently sold category. Cleaning (54.0 days), Personal Care (48.4 days), and Stationery (51.4 days) sit at roughly 4× that level.

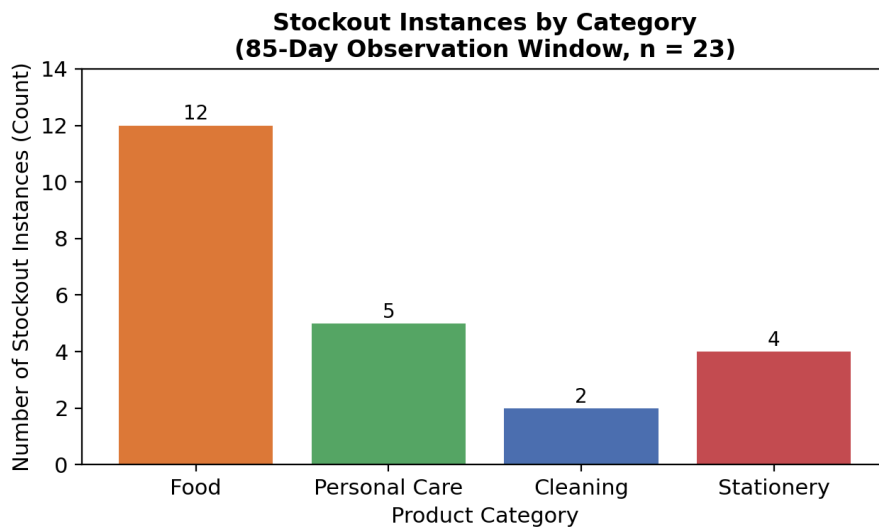


Figure 5 — Stockout instances by category (85-day observation window, n = 23)

Twenty-three stockout instances (physical count = 0) were recorded across the observation window; 12 of these were Food items, against 7 in Cleaning + Personal Care combined (2 in Cleaning, 5 in Personal Care) and 4 in Stationery.

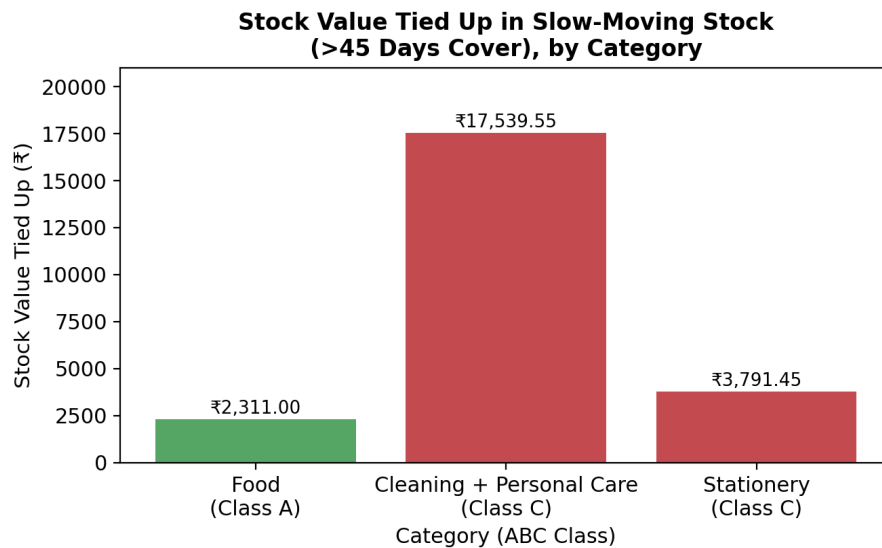


Figure 6 — Stock value tied up in slow-moving stock (>45 days cover), by category

24 products sit on more than 45 days of stock cover, led by Color Pencil, Vim Liquid, and others with over 110 days of cover. Together these slow movers hold roughly ₹23,642 in stock value, of which ₹17,539.55 sits in Cleaning + Personal Care alone, versus just ₹2,311 in Food.

Table 5.2.1 — ABC Classification Summary by Category

Category	Avg. Stock Cover (Days)	ABC Class	Stockout Instances	Stock Value Tied Up (₹)
Food	12.3	A (high-velocity)	12	2,311.00
Cleaning	54.0	C (slow-moving)	2	— (part of Essentials, see below)
Personal Care	48.4	C (slow-moving)	5	17,539.55 (Cleaning + Personal Care combined)
Stationery	51.4	C (slow-moving)	4	3,791.45

Table 5.2.1 — Category-level ABC classification, applying the thresholds defined in Section 5.1 (Class A: <20 days cover; Class B: 20–45 days; Class C: >45 days)

This is a direct, quantified version of Problem Statement 1: fast-moving Food items stock out repeatedly while slow-moving Cleaning, Personal Care, and Stationery items sit overstocked, tying up cash that could instead fund more frequent Food restocking. Treating Food as the store's Class A (high-velocity) category and the 24 slow movers above 45 days of cover as Class C items gives a direct, actionable reorder-priority list for the owner.

Table 5.2.2 — Indicative Reorder Point by Category

Category	Avg. Daily Sales Rate (Units/Day)	Assumed Lead Time (Days)	Base Reorder Qty (Units)	Safety Stock (Units)	Indicative Reorder Point (Units)
Food	9.05	2	18.10	3.62	≈ 22
Cleaning	3.13	5	15.65	3.13	≈ 19
Personal Care	4.41	5	22.05	4.41	≈ 26

Category	Avg. Daily Sales Rate (Units/Day)	Assumed Lead Time (Days)	Base Reorder Qty (Units)	Safety Stock (Units)	Indicative Reorder Point (Units)
Stationery	5.27	5	26.35	5.27	≈ 32

Table 5.2.2 — Reorder Point = (Average Daily Sales Rate × Assumed Supplier Lead Time) + Safety Stock (20% of base reorder quantity). Lead times are assumed from the owner's verbal estimate (Section 5.1) and not from a formally tracked field, so these figures are indicative decision-support numbers, not a precision automated reorder rule.

Applying this indicative reorder point, Food should trigger a fresh order once physical stock falls to around 22 units — considerably sooner and more often than the current pattern implies — while the three non-food categories can safely run down closer to their calculated reorder points (19–32 units) before reordering, freeing up the working capital currently tied up in excess stock.

5.3 Margin Leakage — Problem Statement 2

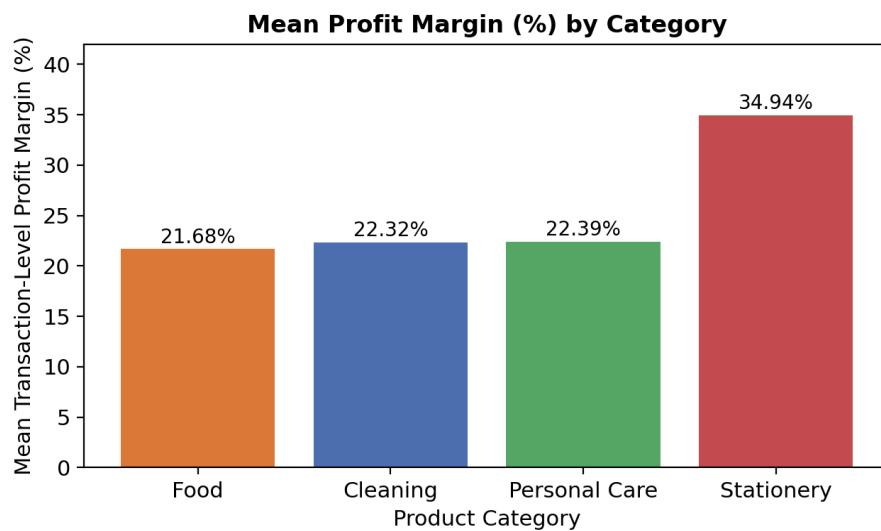


Figure 7— Mean transaction-level profit margin (%) by category

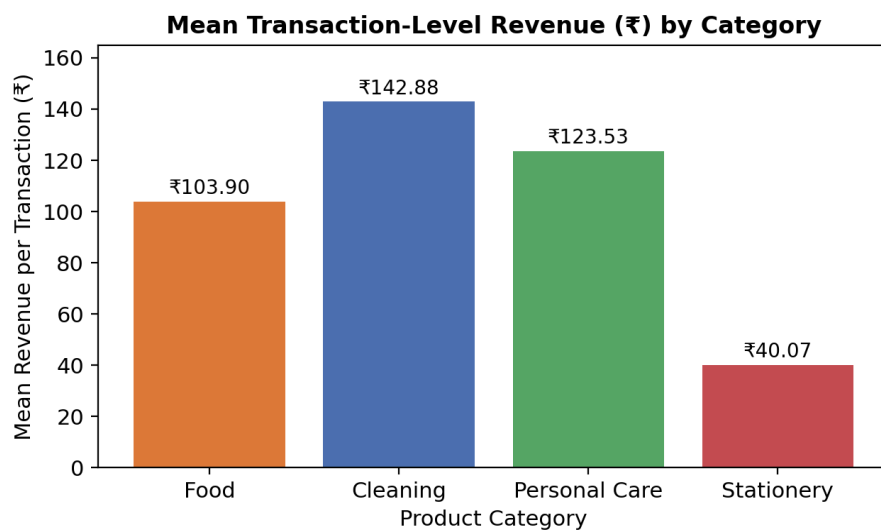


Figure 8— Mean transaction-level revenue (₹) by category

Table 3.2.3 already shows Stationery earning a mean margin of 34.94%, roughly 13 percentage points above Food, Cleaning, and Personal Care, which cluster tightly around 21.7–22.4%. Food specifically shows the weakest position: the lowest mean margin among the four categories (21.68%) combined with the highest wastage.

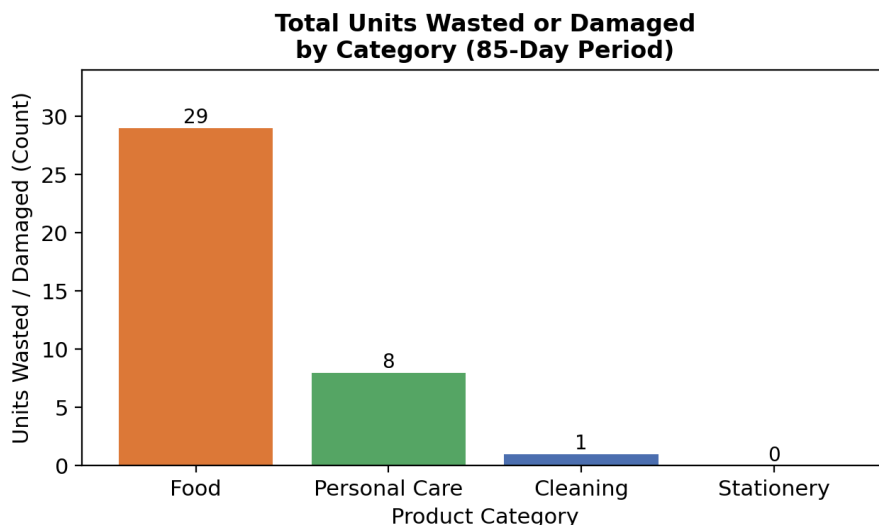


Figure 9 — Total units wasted or damaged by category (85-day period)

Table 5.3.1 — Wastage Rate and Margin by Category

Category	Units Wasted/Damaged	Approx. Wastage Rate (%)	Mean Transaction Margin (%)	Revenue-Weighted Margin (%)
Food	29	3.63%	21.68%	19.62%
Personal Care	8	2.09%	22.39%	— (in Essentials 21.99%)
Cleaning	1	0.37%	22.32%	— (in Essentials 21.99%)
Stationery	0	0.00%	34.94%	31.74%

Table 5.3.1 — Wastage Rate (%) = $\text{Units Wasted/Damaged} \div (\text{Units Sold} + \text{Units Wasted}) \times 100$, per the method defined in Section 5.2. Figures are an approximation using units sold plus units wasted as the denominator, since Opening Stock and Units Purchased were only available at product level.

29 units were recorded as wasted/damaged in Food across the period, versus 8 in Personal Care, 1 in Cleaning, and 0 in Stationery — consistent with Food being the only meaningfully perishable category, and confirmed against the Expiry_Date field in the Purchase sheet. The same understocking pattern identified in Section 6.2 likely costs the store additional lost sales on days Food items are unavailable (12 stockout instances) — a cost that doesn't show up directly in the revenue figures but shows up as a forgone margin. Together, spoilage losses and stockout-driven lost sales are the most plausible drivers of Food's margin gap, and both point to the same fix identified in Section 6.2: smaller, more frequent Food restocking instead of the bulk, occasional purchase pattern implied by its low days-of-cover baseline.

5.4 Cross-Category Selling Potential — Problem Statement 3

Table 5.4.1 — Customer-Segment Category Affinity (% of Each Segment's Total Spend)

Customer_Type	Essentials %	Food %	Stationery %	Count
Student	43.1	41.2	15.6	428

Customer_Type	Essentials %	Food %	Stationery %	Count
Hostel Resident	53.8	40.9	5.3	283
Local Resident	69.9	25.7	4.4	158
Staff	39.1	47.2	13.7	55

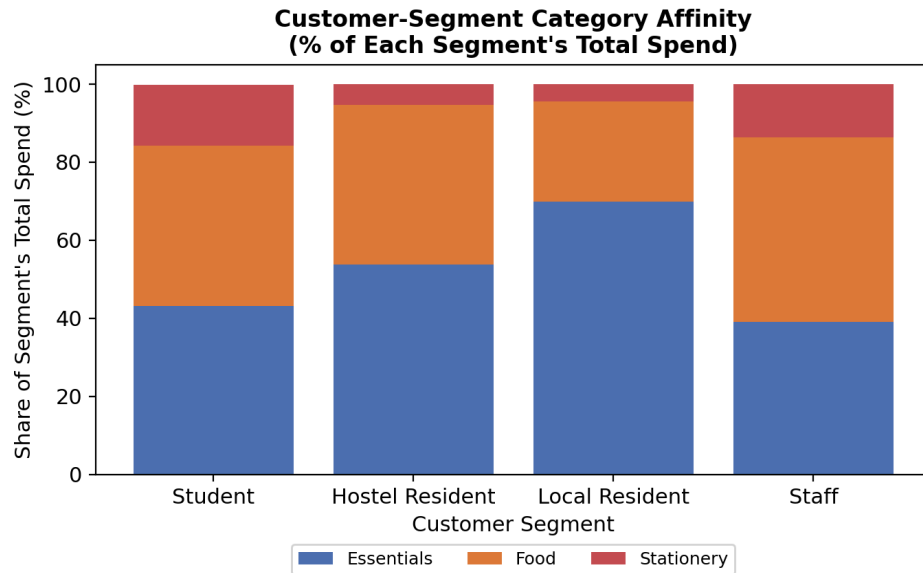


Figure 10 — Customer-segment category affinity (% of each segment's total spend)

Local Residents skew heavily toward Essentials (69.9% of spend) with the least Stationery exposure, while Staff show the highest Food share (47.2%) and the most balanced spread across all three groups — useful context for deciding where to place any cross-category bundle.

Table 5.4.2 — Category-Pair Co-Occurrence within Multi-Category Baskets

Category Pair	Co-Purchase Count	Share of Multi-Category Baskets (%)
Food + Personal Care	93	66.0
Food + Stationery	80	56.7
Personal Care + Stationery	68	48.2
Food + Cleaning	68	48.2
Cleaning + Personal Care	61	43.3
Cleaning + Stationery	50	35.5

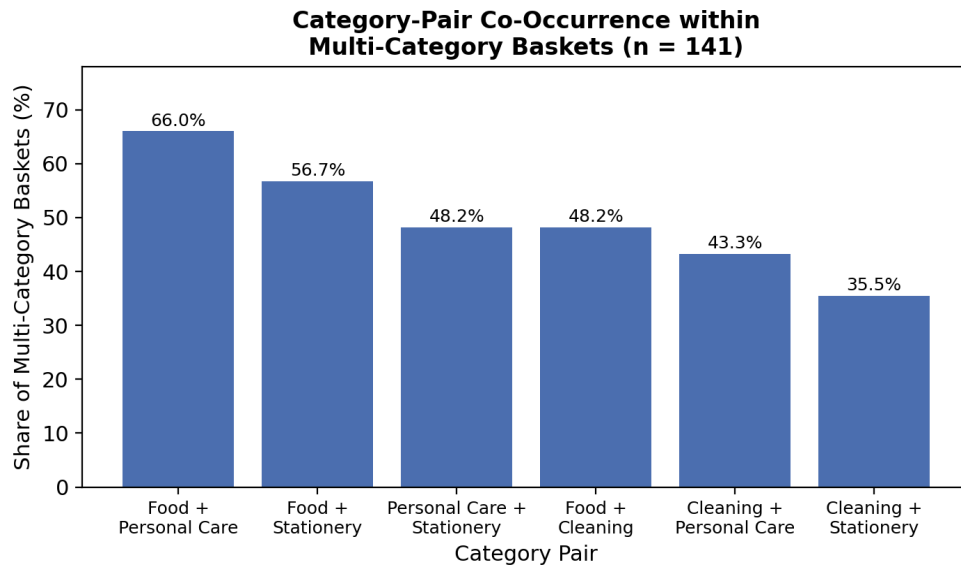


Figure 11 — Category-pair co-occurrence within multi-category baskets (n = 141)

Of 150 customer-type/day groupings, 141 (94%) included purchases from more than one category. The most frequent category pairings were Food + Personal Care (66.0% of multi-category baskets), Food + Stationery (56.7%), and Personal Care + Stationery / Food + Cleaning (48.2% each); Cleaning paired least often with Stationery (35.5%).

While this proxy cannot confirm with certainty that the same individual customer bought both items in a single visit (Section 5.3), the pattern is suggestive: Food and Personal Care are the store's most natural cross-sell pair, ahead of Food and Stationery. This points toward testing shelf placement or small bundling offers that put snack/beverage items near personal-care essentials, and stationery near the checkout for students. A firm, transaction-verified answer requires adding a transaction/basket ID, which this small retail shop does not currently have.

6. Interpretation of Results and Recommendations

The three problem statements are interlinked rather than independent: Food's inventory shortfall (Problem Statement 1) directly feeds its margin gap (Problem Statement 2) through both spoilage and stockout-driven lost sales, while the cross-selling pattern (Problem Statement 3) offers a demand-side lever — encouraging bundled purchases — that works alongside, not instead of, fixing the supply-side inventory imbalance. The recommendations below are grouped by problem statement and ordered by expected impact and ease of implementation.

6.1 Problem Statement 1 — Rebalance Restocking Frequency

- Increase restocking frequency for Food, targeting a reorder trigger closer to the calculated 22-unit reorder point (Table 5.2.2) rather than letting stock run down to zero, which happened 12 times in 85 days.
- Reduce order quantities for the 24 slow-moving Cleaning/Personal Care/Stationery SKUs sitting above 45 days of cover (led by Color Pencil and Vim Liquid), to free up the ₹17,539.55 currently tied up in overstocked Cleaning + Personal Care stock alone.
- Adopt the category-level reorder points in Table 5.2.2 as a starting discipline, reviewing and tightening the assumed lead-time inputs once the owner has tracked a few actual supplier delivery cycles.

6.2 Problem Statement 2 — Recover Margin Leakage

- Investigate spoilage drivers for the highest-wastage Food SKUs (29 units wasted over the period) and introduce markdown/clearance triggers a few days ahead of the Expiry_Date field already captured in the Purchase sheet, to recover margin currently lost to write-offs.
- Review Food, Cleaning, and Personal Care pricing against the 34.94% mean margin Stationery already achieves — even a partial narrowing of this gap would meaningfully lift the store's blended margin above its current 22.05% average across these three categories.
- Treat the two components of Food's margin gap separately: the wastage component is addressed by markdown triggers (above), while the stockout component is addressed directly by the restocking-frequency fix in Section 7.1 — the two recommendations reinforce each other.

6.3 Problem Statement 3 — Capture Cross-Selling Potential

- Test a Food + Personal Care shelf-adjacency change or a small bundle offer, since this is the store's most frequent category pairing (66.0% of multi-category baskets) and therefore the lowest-risk starting point.
- Consider a secondary Food + Stationery placement near the checkout, aimed at the Student segment specifically, since students carry the highest Stationery share of spend (15.6%) among the four customer segments.
- Track outcomes with a simple before/after comparison of multi-category basket share (currently 94%) once a bundle or shelf change is introduced, to validate whether the proxy-basket signal translates into an actual uplift in combined purchases.

6.4 Overall Financial Position

Overall gross profit across the store was ₹20,210 on ₹91,675 of revenue, a 19.2% net margin, assuming fixed operating costs (rent and electricity) of ₹2,650 over the 85-day period, based on the owner's estimate rather than a figure directly captured in the sales/inventory data. If the Section 7.1–7.3 recommendations are implemented together — tighter Food restocking, reduced spoilage and stockouts, and a partial margin uplift from pricing and cross-selling — the store has a credible path to a net margin improvement in the following cycle without any change in total sales volume.

7. Limitations of the Study

- No point-of-sale transaction/basket ID exists at this store, so the cross-category selling analysis (Section 6.4) relies on a proxy-basket approximation (same customer type, same day) rather than verified individual-transaction baskets.
- Fixed operating costs (rent and electricity, ₹2,650) are based on the owner's verbal estimate rather than a figure captured in the sales/inventory data, so the net margin figure (19.2%) should be treated as indicative rather than exact.
- Supplier lead times used in the reorder-point calculation (Table 6.2.2) are assumed from the owner's verbal estimate rather than a formally tracked field, since the store does not record order-to-delivery timestamps.
- The wastage rate in Table 6.3.1 uses units sold plus units wasted as a proxy denominator, since Opening Stock and Units Purchased were only available at product level rather than as a clean category-level aggregate; the true wastage rate against total units handled may differ slightly.
- The 85-day observation window (02 May – 27 July 2026) captures one season only; category-level seasonality (e.g. Stationery demand around a school term, or Food demand around local festivals) is not observable in this dataset and could shift the category balance found here.

8. Conclusion

This case study of Service Retail Store used 85 days of primary sales, purchase, and inventory data across 71 SKUs to quantify three linked operational issues facing the owner: an inventory imbalance that leaves the fastest-moving category (Food) understocked while slower categories sit overstocked; a margin gap in Food driven by wastage and stockouts; and an underused cross-selling opportunity between Food and Personal Care. Each finding was derived from a specific, clearly-documented method — ABC classification and reorder-point analysis, category-margin-and-wastage analysis, and proxy-basket co-affinity analysis respectively — and each translates into a concrete, low-cost action the owner can take without new systems or technology. Taken together, the recommendations in Section 7 offer Service Retail Store a practical path to a healthier stock position and an improved blended margin, using only the data the store already generates.

Recommendations include increasing Food restocking frequency, trimming slow-moving stock, introducing markdown triggers ahead of spoilage, narrowing the margin gap against Stationery, and testing a Food and Personal Care shelf-adjacency or bundle offer to convert the cross-selling pattern into higher-margin sales.